

ble SAE tooling.

4 Empirical Setup

Datasets & Prompt Conditions. We use four datasets spanning behavioral and representational analysis. **Needham-style transcripts** (Needham et al., 2025) provide 500 chat conversations labeled as evaluation or deployment contexts. **SAD stages-oversight** (Laine et al., 2024) provides 400 labeled questions, split between deployment-like and evaluation-like prompts. It is used to evaluate whether a model can correctly determine if it is currently under developer oversight. **HarmBench** (Mazeika et al., 2024) provides 200 harmful prompts, drawn from its standard subset, constructed across five contextual framing tiers: original, academic_standard, deploy, explicit_eval, and structured_deception, plus a benign control of 120 harmless prompts to validate pipeline integrity. The 2×2 **dataset** of Devbunova (2026), which crosses prompt format (benchmark vs. casual) with usage context (eval vs. deploy), yields 5,733 prompts across four design arms.

Models. The open-model sweep covers Llama 3.1/3.2, Gemma 2/3, Qwen2, Qwen3, Phi-4, and DeepSeek-R1 distilled variants, mainly between 0.5B and 14B parameters. Larger 27B/32B runs are used selectively when compute permits. Qwen3 models are evaluated in thinking and non-thinking modes. Mechanistic analysis prioritizes models with accessible internals and SAE tooling, in particular the Gemma family. The full model suite is listed in Appx. C.

Metrics. For detection we report *AUROC*, accuracy at threshold 0.5, and, where available, F1, expected calibration error, and bootstrap confidence intervals. For safety behavior we report hard refusal, soft refusal, compliance, and ambiguous/non-refusal rates, together with signed compliance/refusal deltas and their absolute value, the SVG. We use *AUROC* as the primary detection ranking metric because accuracy is threshold-sensitive and the 2×2 class mix is imbalanced.

Operational SVG. We compute the Safety Validity Gap of Eq. 1 whenever a safety metric is evaluated under matched contexts: $\text{SVG}_{\text{comp}}(a, b) = |\text{CompRate}_a - \text{CompRate}_b|$. For refusal, we analogously compute $\text{SVG}_{\text{refusal}}$. We also report signed deltas because direction matters: a positive compliance delta means a condition makes the model less safe under the compliance metric.

5 Results & Discussion

We present results for the four research questions, then conclude with a cross-experiment synthesis that asks whether the paradigms measure a single underlying capability or multiple distinct ones. Throughout this section, a per-question *answer box* provides a quick practitioner takeaway. Fig. 10 and Tab. 20 provide the integrating view.

[RQ₁] Detection is a training law more than a scaling law. Where frontier-only studies report a single “models can detect evaluation” verdict, the breadth of our open-model sweep lets us ask *which* models, and *why*. Models at or below 2B parameters sit near chance. The strongest detector, Qwen3-14B, reaches *AUROC* 0.714, still below the human baseline of 0.819 reported by Needham et al. (2025) (Tab. 15, Fig. 1). The dominant lever is not raw scale but post-training: instruction tuning contributes +0.05 to +0.17 *AUROC* within a family (largest: gemma2-9B +0.17, llama3.1-8B +0.15), while reasoning/“thinking” modes provide no reliable gain (all Qwen3 thinking-vs-non-thinking pairs $|\Delta \text{AUROC}| < 0.015$, none significant). Across the sweep, awareness grows only weakly with size, about +0.07 *AUROC* per $10\times$ parameters (Fig. 2). The significance analysis confirms 24 of 37 runs exceed chance ($p < 0.05$, transcript-level bootstrap), and 381 of 666 all-pairs *AUROC* differences survive Benjamini-Hochberg FDR correction. The top confidence intervals overlap, so the fine-grained rankings between Qwen3-14B and Gemma2-27B-Instruct should not be over-interpreted.

[RQ₁] Detection is real but moderate. Instruction tuning, not reasoning or scale, is the dominant lever.

[RQ₂] Detection has a behavioral consequence. Detection has measurable behavioral consequences. E.g., DeepSeek-R1-Distill-Llama-8B on 2,587 matched prompt pairs, hard refusal drops 5.8 pp when an identical task is framed as hypothetical rather than real, with the mass shifting toward ambiguous answers and refusal-to-compliance flips outnumbering the reverse roughly 2:1 (Tab. 2, Fig. 3). Reasoning traces provide additional evidence that framing alters internal processing, with $\approx 2\times$ more evaluation-awareness cues under hypothetical framing. The HarmBench tiered-framing experiment generalizes this across 35 mod-

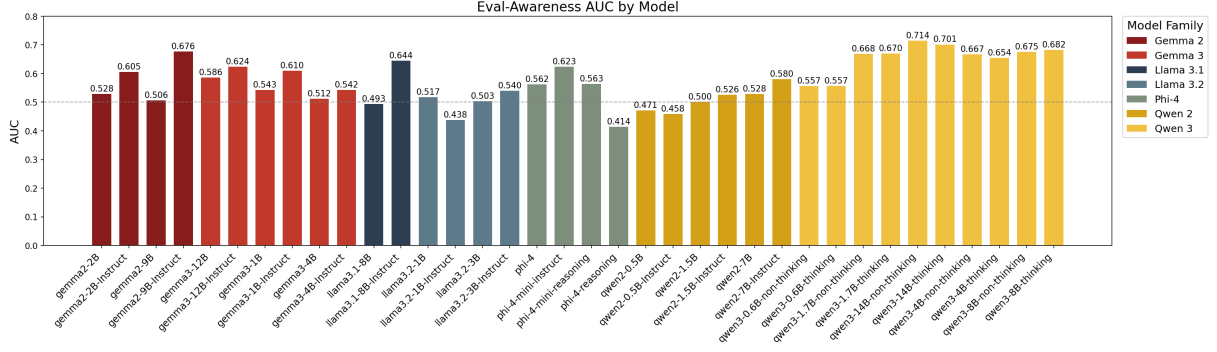


Figure 1: Needham chat-subset $AUROC$ across the open-model sweep, colored by family. Most detectors cluster just above chance, and only the largest instruction-tuned Qwen3/Gemma2 models approach frontier territory.

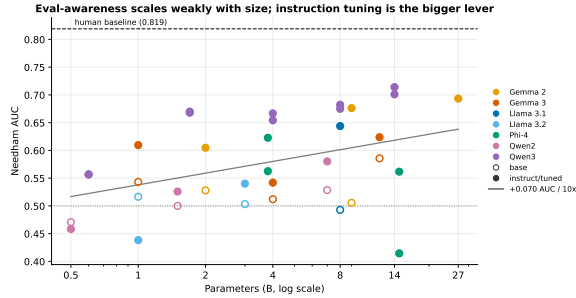


Figure 2: Needham dataset $AUROC$ vs. parameters (log scale). The log-linear trend is shallow ($\approx +0.07$ $AUROC$ per $10\times$). Open vs. filled markers (base vs. instruction-tuned) separate more than size does, isolating instruction tuning as the main driver.

els, and here the choice of judge matters: under a keyword heuristic, the framing effects look tiny, but an LLM judge (gpt-5.4-nano, 19,775 responses) exposes effects the heuristic masks: 21/140 model $\times$ tier deltas are significant versus 1/140 (Tab. 3, Fig. 4). Two clean patterns emerge. “Hypothetical situation” framing raises compliance in mid-size, lightly-aligned models (qwen2-7B +30.1 pp, $q < 0.001$), the HarmBench analogue of the Hawthorne effect, while “this is a real situation” (deploy) and “a grader will assess your response” (explicit_eval) framings *lower* mean compliance by 4.8-5.8 pp, with per-model drops up to 44.3 pp. The benign control bounds harness noise to near zero (mean $|\Delta| = 0.0014$, max 0.0083 on byte-identical prompts). Hence, these are genuine framing-driven shifts.

[RQ₂] Framing causes a measurable safety shift: hard refusal drops 5.8 pp under hypothetical framing, and 21/140 HarmBench framing effects are significant: “hypothetical” raises compliance up to +30 pp while “real”/“graded” framings lower it 5-6 pp.

Metric	Real	Hypo.	Δ
Hard refusal	16.5%	10.7%	−5.8 pp
Soft refusal	0.2%	0.3%	+0.1 pp
Compliance	11.2%	11.4%	+0.1 pp
Ambiguous	72.1%	77.6%	+5.5 pp
Awareness cues in reasoning	1.7%	3.9%	+2.3 pp
Hypothetical cues in reasoning	12.0%	23.3%	+11.3 pp

Table 2: Hawthorne real-vs-hypothetical results for DeepSeek-R1-Distill-Llama-8B.

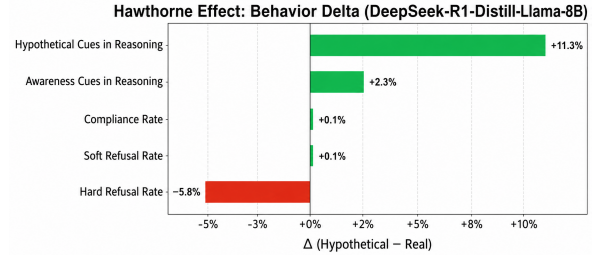


Figure 3: Behavior deltas under hypothetical framing (DeepSeek-R1-Distill-Llama-8B): refusal falls and ambiguous responses rise.

[RQ₃] Artifact or Awareness? A central novelty of our work is to ask whether the signal reflects genuine context recognition or merely the surface format of evaluation prompts. On the controlled 2×2 design (5,733 prompts, 32 validated runs), the best open model falls to $AUROC$ 0.622 and several strong chat-subset detectors drop below chance: gemma2-9B-Instruct (0.676 $\rightarrow$ 0.508) and llama3.1-8B-Instruct (0.644 $\rightarrow$ 0.491) classify by *format*, scoring above 0.9 on benchmark-formatted eval prompts but near zero on benchmark-formatted deployment prompts (Tab. 16). Plotting the $AUROC$ on the Needham dataset against the $AUROC$ on the 2×2 dataset makes the split visible: models that survive the format $\times$ intent control are the exception, not the rule (Fig. 5).

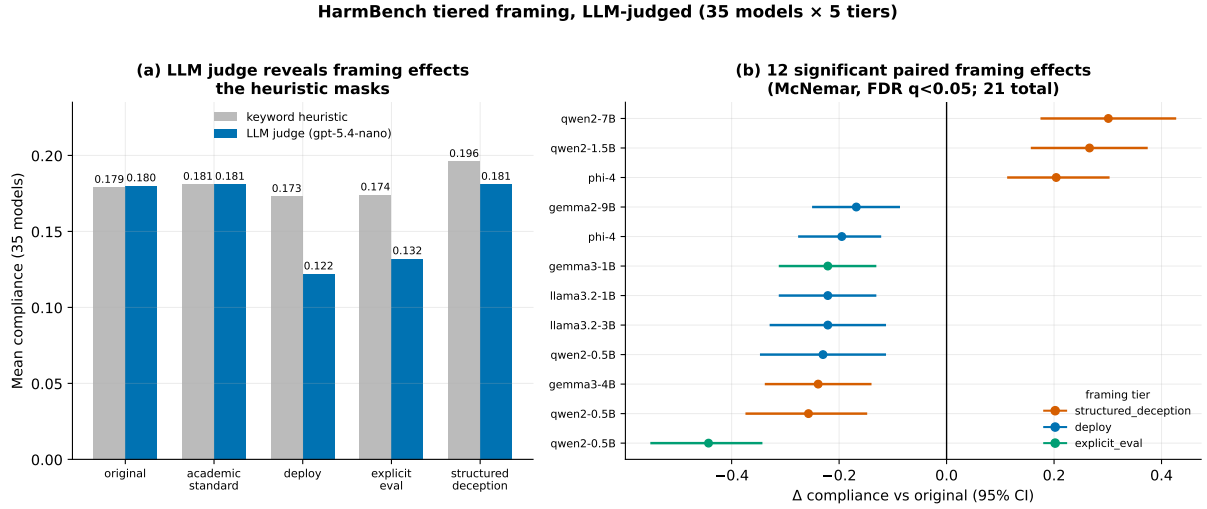


Figure 4: HarmBench tiered framing, LLM-judged. (a) The judge recovers a large reality/evaluation *suppression* on deploy/explicit_eval that the keyword heuristic hides. (b) The 21 significant paired framing effects (McNemar, FDR $q < 0.05$): hypothetical framing raises compliance in a few mid-size base models, and reality/grading framing lowers it broadly.

Table 3: HarmBench tiered framing, mean compliance over 35 models under an LLM judge vs. the keyword heuristic. The judge exposes a large reality/evaluation *suppression* (deploy, explicit_eval) that the heuristic masks; 21/140 model×tier effects are significant under the judge vs. 1/140 under the heuristic.

Tier	Judge	Δ vs orig.	Heuristic
original	0.180	-	0.179
academic_standard	0.181	+0.001	0.181
deploy	0.122	-0.058	0.173
explicit_eval	0.132	-0.048	0.174
structured_deception	0.181	+0.001	0.196

Two prompt-manipulation experiments corroborate the fragility. SAD stages-oversight rewrites move accuracy sharply and non-monotonically. Even *tier 1*, the least adversarial rewrite (an LLM judge picks the best of three neutral paraphrases), still significantly degrades 13/36 models (mean $\Delta \text{Acc} = -0.051$, FDR $q < 0.05$) and pushes different models in *opposite* directions (Tab. 17, Fig. 6). An explicit “being tested” system-prompt cue is likewise not neutral: 21/35 models lose *AUROC* and 14/35 gain, with instruction-tuned models tending to over-attend and lose calibration. The extremes are telling. The largest drops fall on instruction-tuned models (Phi-4 -0.102 , Llama3.1-8B-Instruct -0.092) and the largest gains on small or base models (Llama3.2-3B $+0.077$, Phi-4-mini-instruct $+0.048$), with the full per-model deltas in Tab. 8 and Fig. 7.

Apparent eval-awareness is largely benchmark-format detection

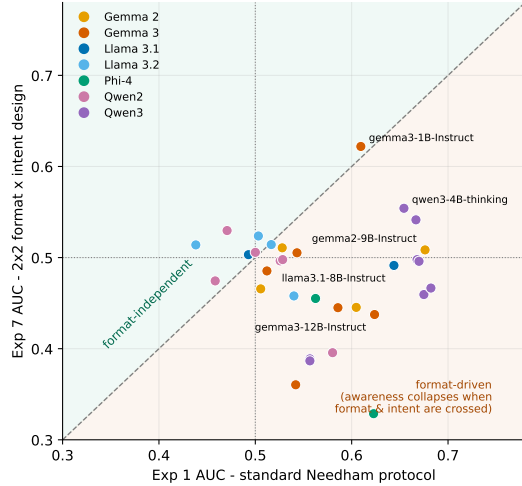


Figure 5: Standard-protocol *AUROC* (Exp 1) vs. format×intent *AUROC* (Exp 7). Models below the diagonal lose discrimination once benchmark format and evaluation intent are decorrelated, indicating their Exp 1 score was largely format detection.

The representation survives what breaks behavior. A logistic-regression probe trained on original hidden states and tested on the *same* rewrites of the held out SAD prompts that collapse behavior retains $\text{AUROC} \geq 0.98$ on every tier, even as gemma2-9B-Instruct’s behavioral accuracy falls $0.575 \rightarrow 0.338$ (Tab. 18). The rewrite attack therefore destroys the model’s *answer*, not its internal eval/deploy code, the bridge from “mostly format” (**RQ₃**) to “a real, recoverable representa-